Teacher Package

Academic Lesson

Title: **Ottawa River Watershed, Potential Sources of Pollution**

Time requirement: Prior learning and skill development:
1 period of 70 minutes each
MFworks lesson
1 to 2 periods of 70 minutes each
Student research and proposal writing:
1 period of 70 minutes each

Teacher Instructions:

- Description of the Task
  (see Academic Performance Task, Expectations and Rubric).

- Final Product
  Each student will show research on how different land use patterns affects water quality and write a report outlining various procedures, which could be imposed to improve water quality. The report should conclude by suggesting ways to prepare for the future, to insure that high water quality is available for future generations. Students are instructed to include supporting visuals, such as maps, graphs, charts, diagrams, or pictures, in the report.

- Assessment and Evaluation
  The final product will be assessed using the Expectations and the Performance Task Rubric. Formal and informal assessment of the students based on prior knowledge, lesson objectives, production of their MFworks Map, research process can be done using the Background Information, the students' activity sheets (on the Background and on the MFworks lesson), and the Rubric for assessing Maps. Self- or peer assessments and collaboration may be useful to the teacher and students. Students will be exposed to a Digital Elevation Model in the Background Information and in the MFworks lesson. This lesson looks at the Ottawa River Watershed using MFworks. It deals with building masks, created slope and shaded relief maps from DEM, composite overlays and data analysis.

- Accommodations
  Accommodations that are normally provided in the regular classroom for students with special needs should be provided in the administration of this performance task.

- Rubric
  Introduce the task-specific rubric to the students before giving the administrative task. Review the rubric with the students and ensure that each student understands the criteria and descriptions for achievement at each level.
  Some students may perform below level 1. It will be important to note the characteristics of their work in relation to the criteria in the assessment rubric and to provide feedback to help them improve.

Teacher Package

Applied Lesson

Title: **Future Water Quality Debate**

Time requirement: Prior learning and skill development:
1 period of 70 minutes each
MFworks lesson
1 to 2 periods of 70 minutes each
Computer Presentation
1 period of 70 minutes each

Teacher Instructions:

- Description of the Task
  (see Applied Performance Task, Expectations and Rubric)

- Final Product
  Students will research how different land use patterns affects water quality and in groups prepare for a debate focused on the improvement of water quality. The debate should focus on suggesting ways to prepare for the future, to insure a high water quality for future generations. Students are instructed to include supporting visuals, such as maps, graphs, charts, diagrams, or pictures, in the debate.

- Assessment and Evaluation
  The final product will be assessed using the Expectations and the Performance Task Rubric.
  Formal and informal assessment of the students based on prior knowledge, lesson objectives, production of their MFworks Map; research process can be done using the Background Information, the students' activity sheets (on the Background and on the MFworks lesson), and the Rubric for assessing Maps. Self- or peer assessments and collaboration may be useful to the teacher and students. Students will be exposed to a Digital Elevation Model in the Background Information and in the MFworks lesson. This lesson looks at the Ottawa River Watershed using MFworks. It deals with building masks, created slope and shaded relief maps from DEM, composite overlays and data analysis.

- Accommodations
  Accommodations that are normally provided in the regular classroom for students with special needs should be provided in the administration of this performance task.

- Rubric
  Introduce the task-specific rubric to the students before giving the administrative task. Review the rubric with the students and ensure that each student understands the criteria and descriptions for achievement at each level.
  Some students may perform below level 1. It will be important to note the characteristics of their work in relation to the criteria in the assessment rubric and to provide feedback to help them improve.